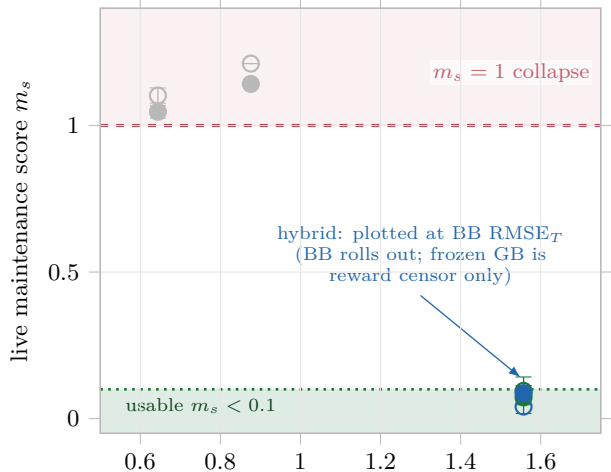
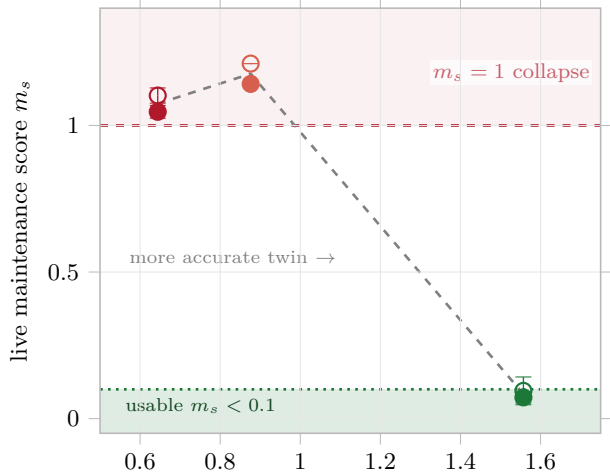


(A) Single-model paradox: more accurate $\rightarrow$ worse live m_s (B) Role-separated hybrid recovers usable control



24 h rollout RMSE_T (°C) ($\leftarrow$ more predictive fidelity) 24 h rollout RMSE_T (°C) ($\leftarrow$ more predictive fidelity)

● BB (hourly) ● matched-BB (15 min) ● GB (calibrated) ● hybrid ● peak ○ typical